

PORTFOLIO PROJECT

CyberAudit & Solutions

SME Audit Management Platform

Stage 3: Technical Documentation

Holberton School Dijon | 2026

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Introduction

1.1 Context

CyberAudit & Solutions is a full-stack web platform that connects French SMEs/VSEs without an internal IT team with cybersecurity experts. According to ANSSI 2023, 45% of SMEs have no dedicated IT team and remain exposed to cyberattacks (ransomware, phishing, data breaches).

1.2 Scope of the document

This technical documentation covers the detailed design of the MVP. It describes:

- User needs formalized into prioritized User Stories
- The software architecture and data flow; the data model (relational PostgreSQL / MySQL)
- The back-end classes, front-end components, interaction scenarios, and exposed and consumed APIs
- The code management and quality strategy

1.3 Target audience

- Development team (Tommy + James): reference during the coding phase (W2 → W12).
- Holberton Jury: proof of the detailed MVP design.
- RNCP 5 Jury: reusable skills portfolio for Web and Mobile Web Developer certification.

1.4 Stack technique

Layer	Technology	Justification
Front-End	React + Tailwind CSS	High-performance SPA, rapid design system
Back-End	Django + Django REST Framework (DRF)	Native security, mature ecosystem
Database	PostgreSQL or MySQL	ACID relational analysis, high-performance indexes
Async	Celery + Redis	Long tasks to be developed beyond the RNCP level 5 web and mobile web developer
Auth	SimpleJWT	Stateless, scalable
Accommodation	Vercel + Railway	HTTPS auto, déploiement Git

User Stories & Mockups

2.1 Personas

Marie: 45 years old, manager of an accounting firm in Dijon, no IT knowledge. UX target < 3 min.

Karim: 32 years old, cybersecurity expert, manages the request queue in parallel.

2.2 User Storie priorisée

User Story	Stage W (Week)
As a visitor, I want to register in order to create my account.	W3-4
As a user, I want to log in to access my account.	W4
As a user, I want to log out in order to protect my account.	W5
As an SME customer, I want to submit a request in < 3 min.	W6
As a customer, I want to track the status (4 statuses).	W7
As a customer, I want to receive an email for every change.	W8
As an admin, I want to see all requests sorted.	W9
As an admin, I want to change the status.	W10
As an admin, I want to generate a PDF.	W10
As a customer, I want to download my PDF.	W10

Should Have

- Awareness modules
- Password reset
- Admin filters
- UI notifications.

Could Have / Won't Have

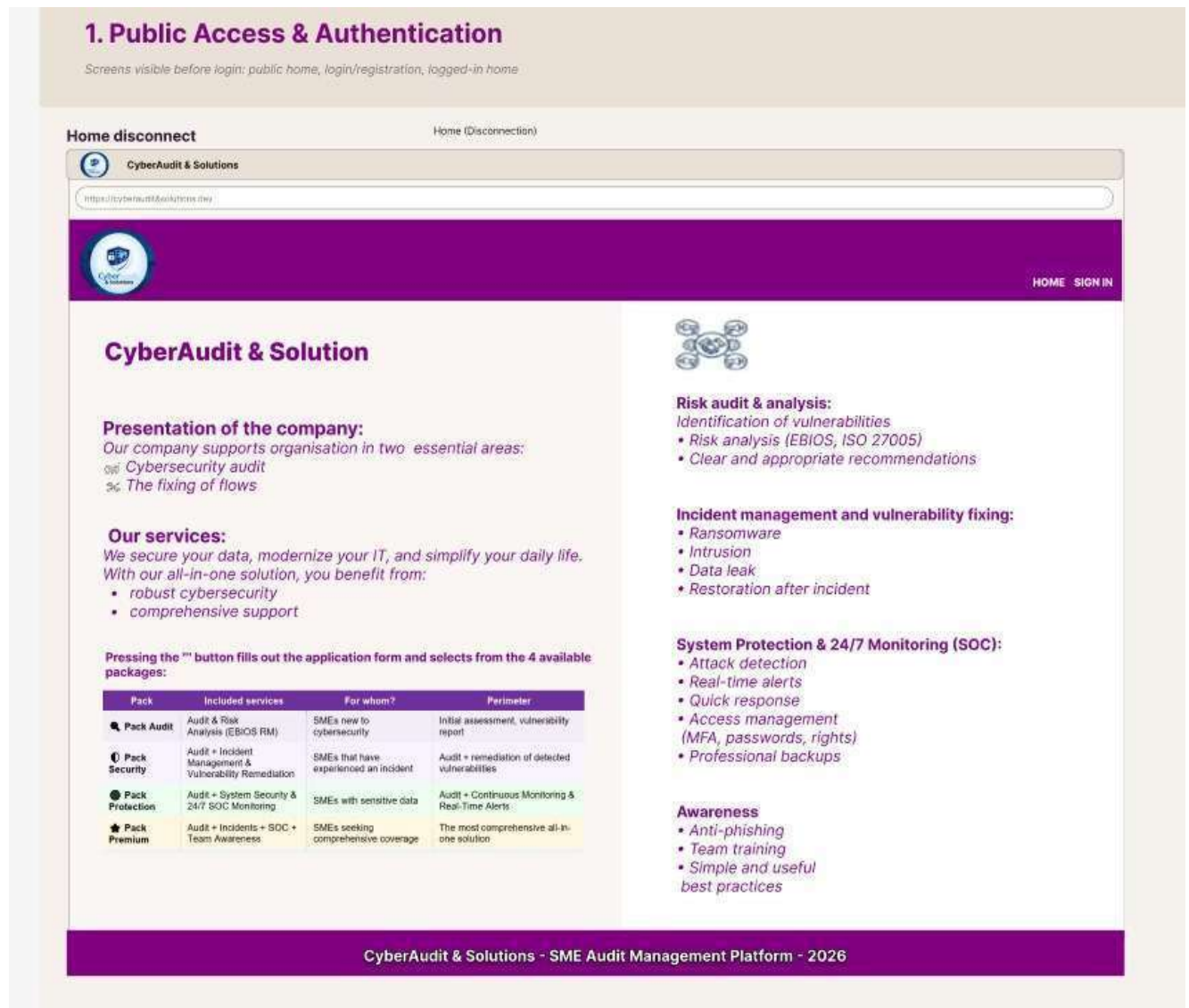
Outside the scope:

- mobile app
- AI scoring
- Marketplace
- OAuth2
- gamification

2.3 Screen Descriptions (11 Figma mockups to be produced in W1)

Screen 1: Public Landing

The CyberAudit & Solutions platform homepage presents the company, its services, and the packages offered, tailored to entrepreneurs with fewer than 50 employees. Simply go to the "SIGN IN" tab so the client (Marie) can create an account and log in.



Screen 2: Registration

After clicking "Create an account," the customer enters her company name, first name, last name, email address, and password (hashed).

Form completion rules:

- Each field contains a maximum of 50 characters.
- If a field is missing error message "Please enter the information on the form."
- Email address format: name@domain required.
- Password: one lowercase letter, one uppercase letter, one number, one special character, minimum 10 characters.

The screenshot displays the 'Create an account' registration form within a web browser. The browser's address bar shows 'http://cyberaudit&solutions.qwr'. The page features a purple header with the 'CyberAudit & Solutions' logo and navigation links for 'HOME' and 'SIGN IN'. The registration form is centered and includes the following fields: 'Company Name' (placeholder: 'Enter company name'), 'First Name' (placeholder: 'Enter first name'), 'Last Name' (placeholder: 'Enter last name'), 'Email' (placeholder: 'Enter email address'), and 'Password' (placeholder: 'Create a password'). A purple 'Create an account' button is positioned below the password field. Below the button, there is a link 'Already have an account?' and a 'Sign In' button. The footer of the page reads 'CyberAudit & Solutions - SME Audit Management Platform - 2026'.

Screen 3: Login

The customer can log in with their account; otherwise, they click on "Create an account."

"Lost Mail" and "Lost Password" links are planned (to be developed later for the presentation RNCP 5).

The screenshot displays the login interface for the CyberAudit & Solutions platform. At the top, a navigation bar includes a "Sign in" link and a "Log in" button. Below this, a browser window shows the URL "https://cyberauditandsolutions.zew". The main header features the company logo and navigation links for "HOME" and "SIGN IN". The central focus is a "Log In" form with a lock icon, prompting users to "Enter your credentials to access your account". The form contains input fields for "Email / User name" and "Password", each with placeholder text. Below these fields are links for "Lost Mail ?" and "Lost Password ?". A prominent purple "Connection" button is positioned below the form. At the bottom of the form, a link "Don't have an account?" is followed by a "Create an account" button. The footer of the page reads "CyberAudit & Solutions - SME Audit Management Platform - 2026".

Screen 4: Client Dashboard

When the client is logged in, she can view her dashboard to track the progress of her audit request (Pending / In Progress / Completed). She can click on "View details" and download the report via "Download report".

Client Dashboard

CyberAudit & Solutions

https://cyberaudit18solutions.dev

HOME SIGN OUT

Client Portal

- Dashboard
- Audit Request
- Training

My Dashboard

Welcome Marie - Real-time tracking of your audit requests

Open requests
2

Completed
1

Reports available
1

My audit requests

Filter: All statuses All packs

Case #	Pack	Submission	Status	Action
DOSSIER-2026-0001	Premium	29/04/2026	In Progress	View details
DOSSIER-2026-0002	Audit	22/04/2026	Pending	View details
DOSSIER-2026-0003	Security	10/04/2026	Completed	Download report

Recent notifications

DOSSIER-2026-0001 - Status changed: Pending → In Progress (2h ago)

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Screen 5: Audit Form

By clicking the "Audit Request" tab, the client accesses the audit request form. She must fill in:

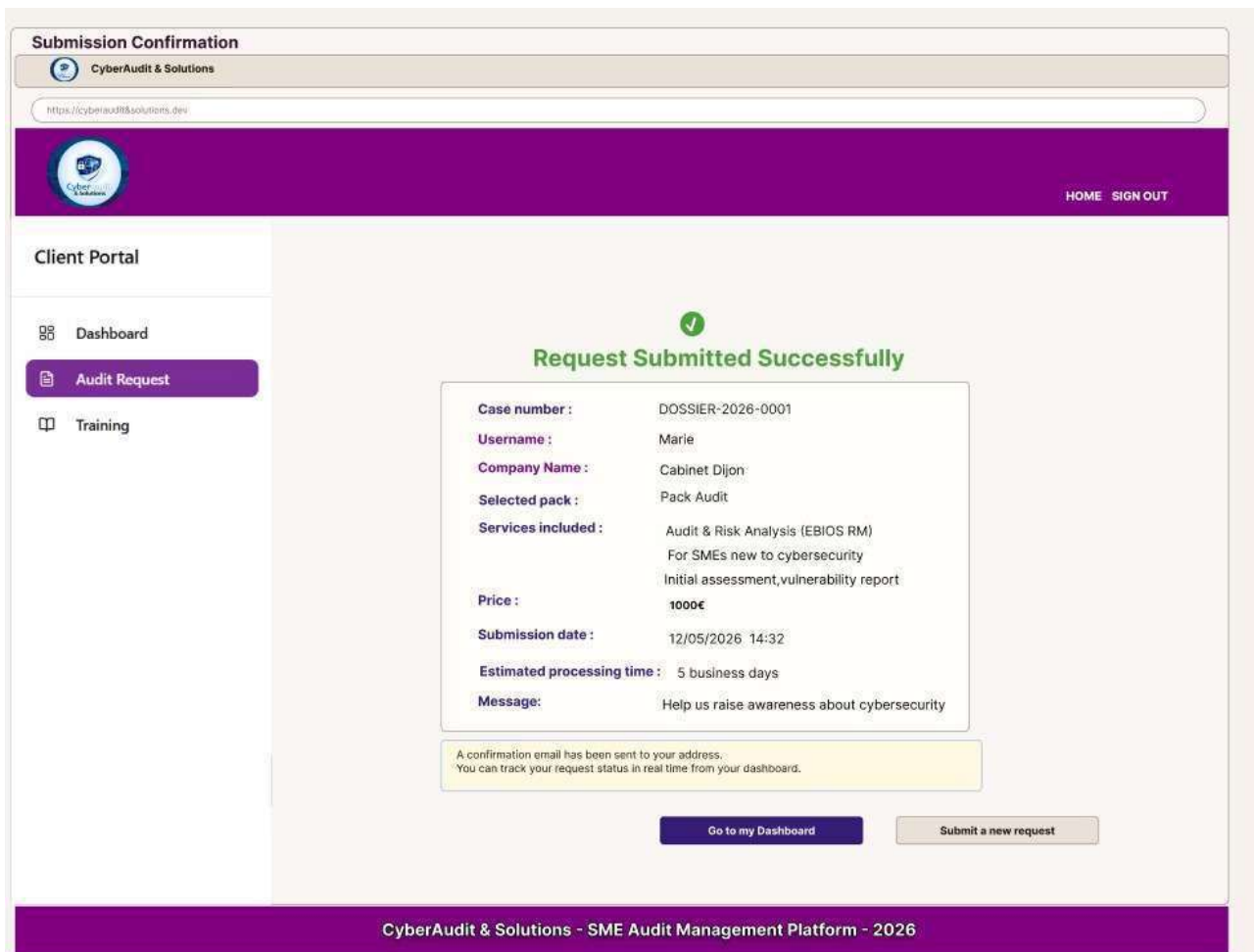
- Her username (maximum 50 characters)
- The company name (less than 50 characters)
- Choose the package (Audit / Security / Protection / Premium). When the client selects the package using the radio button, the package details are displayed in the "Includes Services" panel, including the "Included Services," "For whom and what needs are being addressed?", "Scope of analysis," "Analysis estimate," and "Price."
- An optional message describing the desired audit request.

After completing the form, the client clicks "Send the Audit Request" to submit the audit request.

The screenshot displays the 'CyberAudit & Solutions' Client Portal. The left sidebar contains a 'Client Portal' header and three menu items: 'Dashboard', 'Audit Request' (highlighted), and 'Training'. The main content area is titled 'Audit request form'. It includes input fields for 'Username' and 'Company Name'. Below these is a 'Select pack' section with four radio button options: 'Pack Audit' (selected), 'Pack Security', 'Pack Protection', and 'Pack Premium'. To the right of the radio buttons is an 'Included Services' panel that displays details for the selected 'Pack Audit', including a list of services, a scope of analysis, an analysis estimate, and a price. At the bottom of the form is a 'Message' input field and a 'Send the audit request' button. The footer of the page reads 'CyberAudit & Solutions - SME Audit Management Platform - 2026'.

Screen 6: Request details (client)

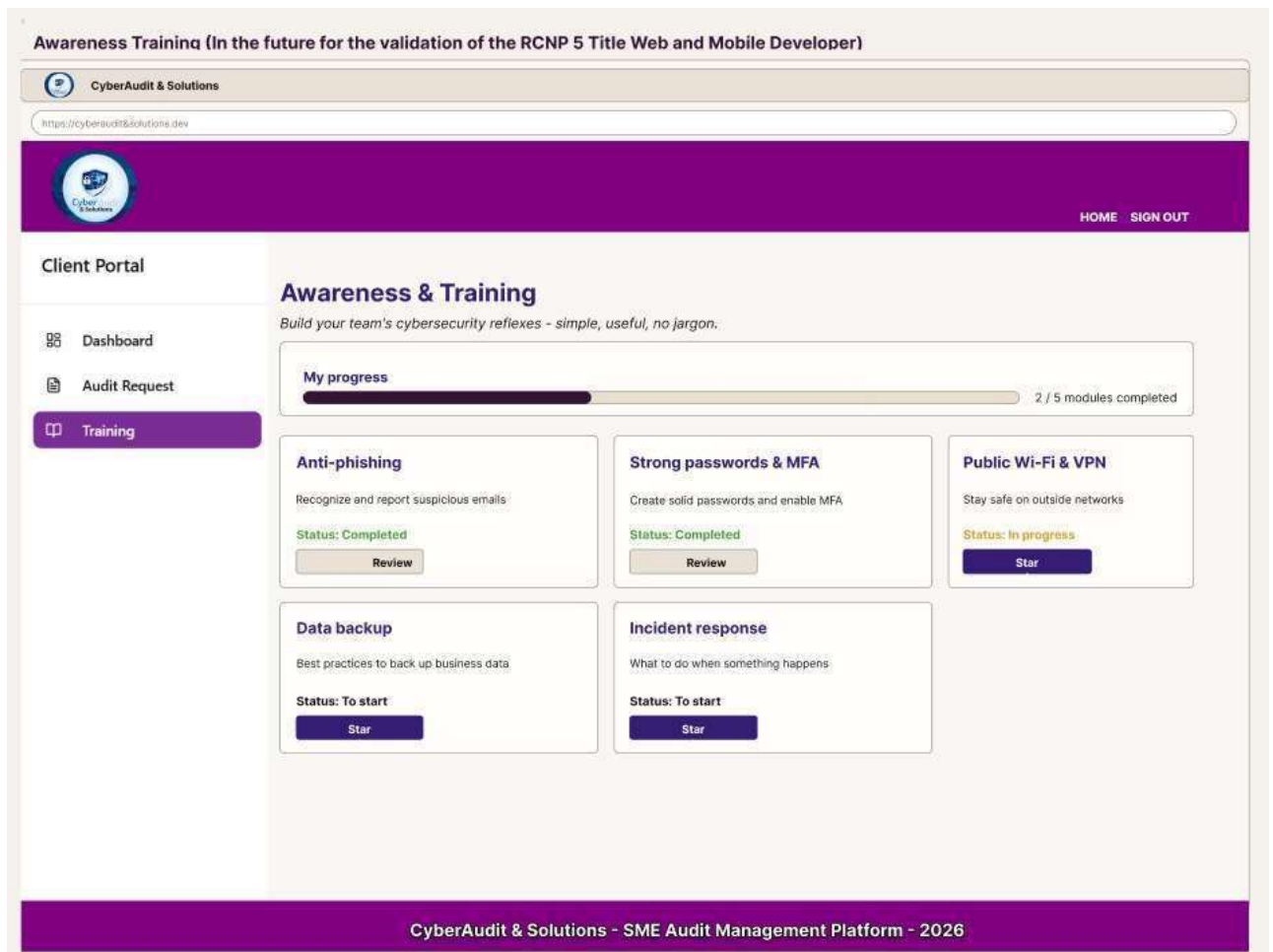
After sending, confirmation message to the expert (Karim) and acknowledgment of receipt to the client. The client follows the progress of the analysis.



Screen 7: Training Module

To be developed later. Modules: 'Anti-phishing', 'Multi-factor Authentication', 'Public Connection and Wi-Fi & VPN', 'Data Backup', 'Incident Response'. In various formats (video, video conference, documentation) with a progress bar.

This part of the section will be presented in the future beyond the RCNP 5 Web and Mobile Developer title. The CyberAudit & Solutions team should help us prepare the training modules in the application.



Screen 8: Admin Dashboard

The expert (Karim) accesses his dashboard with all the requests. He can sort the list according to the chosen package and the current statuses, then view and edit each request.

Admin Dashboard

CyberAudit & Solutions

https://cyberauditsolutions.dev

HOME SIGN OUT

Admin Portal

- Dashboard
- Request
- Report

Admin Dashboard - Audit Requests

CyberAudit operator view - Manage all SME requests

Pending: 5

In Progress: 8

Completed: 23

Archived: 12

Filters :

Status : All Pack : All Date : DD/MM/YYYY Client : Search Apply Reset filters

Case #	Client	Pack	Submitted	Status	Action
DOSSIER-2026-0001	Marie SME	Premium	29/04/2026	In Progress	View / Edit
DOSSIER-2026-0002	Tech Corp	Audit	22/04/2026	Pending	View / Edit
DOSSIER-2026-0003	BlueRoot SARL	Security	10/04/2026	Completed	View report
DOSSIER-2026-0004	City Hall Dijon	Protection	03/04/2026	In Progress	View / Edit
DOSSIER-2026-0005	Atelier Pro	Audit	25/03/2026	Archived	View report

5 of 48 requests

1 2 3 4 ..

CyberAudit & Solutions - SME Audit Management Platform - 2026

Screen 9: Request Detail (admin)

By clicking on "View/edit," the expert can view the client's request, change the status via the "Status & Actions" form, generate a PDF report via "Generate PDF report," send a notification to the client ("Send notification"), and can archive the request ("Archive request").

Admin Request Detail

CyberAudit & Solutions

https://cyberauditsolutions.dev



HOME SIGN OUT

Admin Portal

Dashboard

Request

Report

Request DOSSIER-2026-0001

Back to list

Client information

Company : Marie SME (Cabinet Comptable)

Contact : Marie Dupont

Email : marie@cabinet-dijon.fr

Submitted : 29/04/2026 at 14:32

Pack selected : Pack Audit

Client message : Help us raise awareness about cybersecurity

Status & Actions

Current status : In Progress

Assigned to : Karim

Internal notes

EBIOS RMI analysis ongoing. Awaiting client log files.

Save changes

Generate PDF report

Send notification

Archive request

History (immutable - GDPR)

Date	Author	Action
29/04/2026 14:32	Marie Dupont	Request submitted (Pack Premium)
29/04/2026 14:32	System	Confirmation email sent (Celery)
29/04/2026 16:10	Karim	Status changed: Pending → In Progress
29/04/2026 16:11	Karim	Internal note added

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Screen 10: PDF report viewer

Report can be downloaded. The report identifies:

- The level of vulnerabilities (Low, Medium, High)
- The assets (mail server, wifi router, etc.)
- The description and the recommendations
- An action plan is generated with the estimated number of days.

PDF Report Viewer

CyberAudit & Solutions

https://cyberaudit&solutions.dev

HOME SIGN OUT

Admin Portal

Dashboard

Request

Report

Vulnerability Report - DOSSIER-2026-0001 [Download PDF](#)

Global security score

B+ Acceptable - 3 medium-severity vulnerabilities found 72 / 100

Executive summary

EBIOS RM analysis performed on the IS perimeter declared by the client. 3 vulnerabilities identified, none critical. Action plan attached, prioritized over 30 days.

Vulnerabilities identified

Severity	Asset	Description	Recommendation
Medium	Email server	No DMARC policy	Activate DMARC + SPF + DKIM
Medium	Wifi router	Default WPA2 password	Rotate password and enable WPA3
Low	Workstation	Outdated antivirus	Update Defender + central console

30-day action plan

Week 1: DMARC/SPF/DKIM activation
 Week 2: Wi-Fi rotation + WPA3 migration
 Week 3: Endpoint update campaign
 Week 4: Final compliance check

CyberAudit & Solutions - SME Audit Management Platform - 2026

Screen 11: Download from the dashboard

The client can download the report from their dashboard via the "Download report" link.

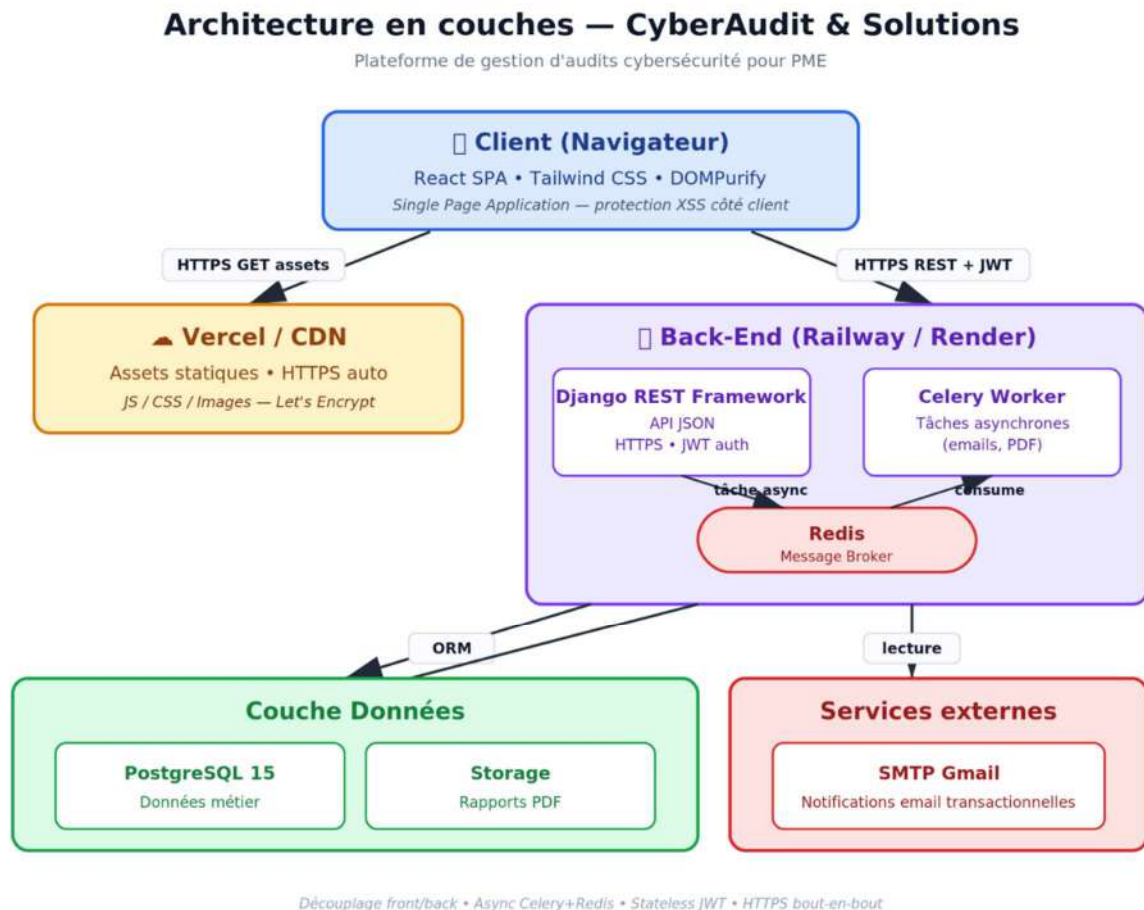
The screenshot displays the 'Client Portal' dashboard. On the left, a sidebar contains links for 'Dashboard', 'Audit Request', and 'Training'. The main content area is titled 'My Dashboard' and includes a welcome message for 'Marie'. It features three summary cards: 'Open requests' (2), 'Completed' (1), and 'Reports available' (1). Below these is a section titled 'My audit requests' with filter dropdowns for 'All statuses' and 'All packs'. A table lists three audit requests, with the 'Download report' link for the 'Completed' status highlighted in yellow.

Case #	Pack	Submission	Status	Action
DOSSIER-2026-0001	Premium	29/04/2026	In Progress	View details
DOSSIER-2026-0002	Audit	22/04/2026	Pending	View details
DOSSIER-2026-0003	Security	10/04/2026	Completed	Download report

System architecture

3.1 Overview: Layered Architecture

The architecture follows a layered model:



Layered architecture:

Client (browser) → CDN/Backend → Data + External Services.

Client (Browser):

- React SPA
- Tailwind CSS
- DOMPurify (anti-XSS)
- HTTPS
- REST + JWT

Vercel / CDN :

- Static assets
- HTTPS Let's Encrypt

Back-End (Railway / Render):

- Django REST API (JSON, JWT, HTTPS)
- Celery Worker (async)
- Redis (broker)
- ORM
- SMTP

Data layer: PostgreSQL or MySQL + PDF Storage.

External services : Gmail SMTP (email notifications).

3.2 Data Flow: Audit Request

Client fills out the form in the React SPA.

The front end sends POST /api/audits/ (JSON + JWT in Authorization: Bearer).

Django validates the request (DRF serializer) and persists it in the database via the ORM.

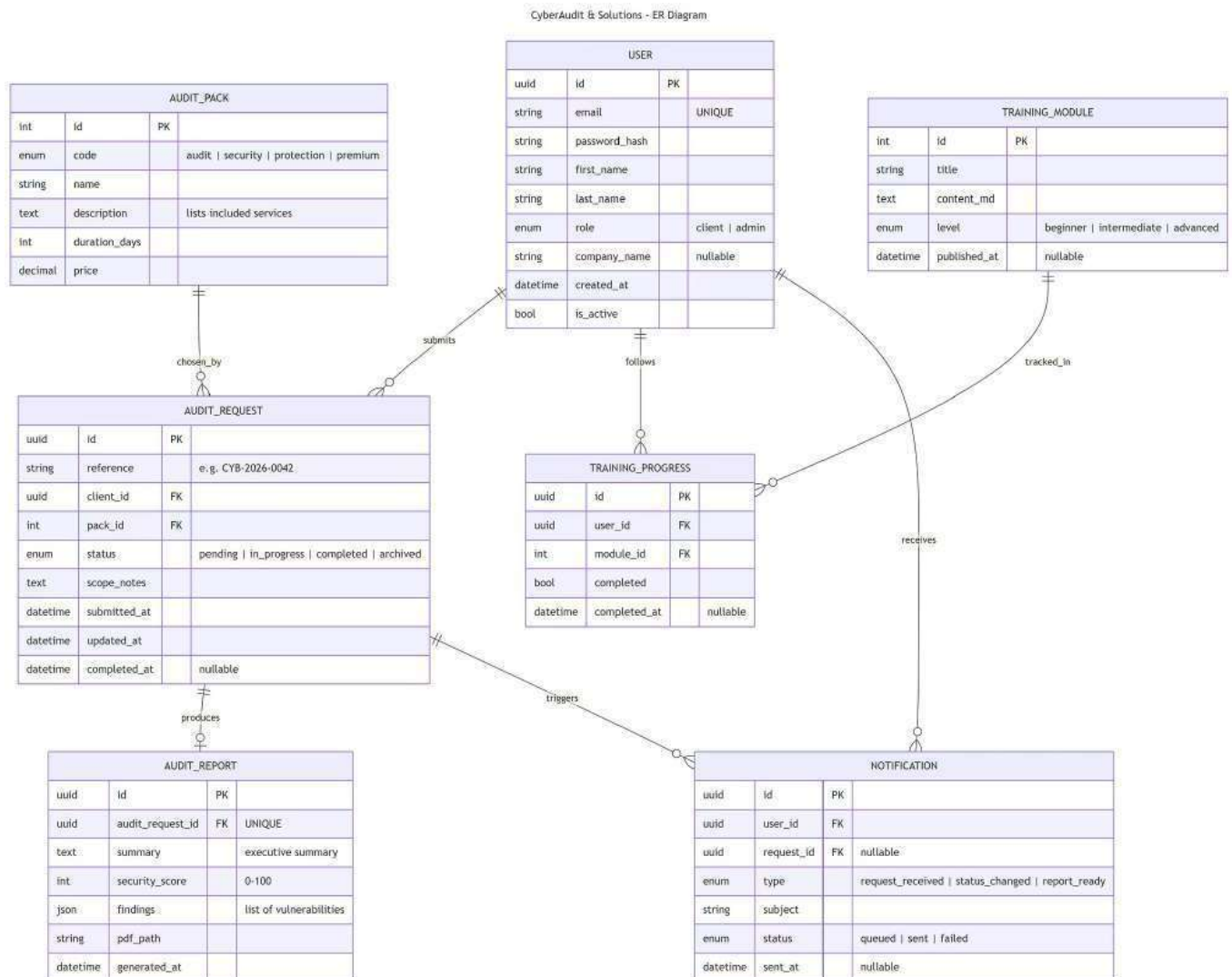
Django publishes a Celery task `send_acknowledgment_email(audit_id)` in Redis.

3.3 Justified architecture choices

- **Front/back decoupling** → 2 independent deployments, controlled CORS, cacheable SPA on CDN.
- **Async Celery** → generating a PDF can take 5-10 s; the HTTP thread is not blocked.
- **PostgreSQL relational** → highly structured data with relationships (User → AuditRequest → Report), critical ACID integrity.
- **JWT stateless** → no server session, horizontally scalable

Components, classes and DB

4.1 Database Schema



Entity	Key fields	Relationship
USER	id (UUID), email (UK), password_hash, first_name, last_name, role (client/admin), company_name, created_at, is_active	1→N AuditRequest (submits / follows / receives)
AUDIT_PACK	id, reference, code (audit/security/protection/premium), name, description, duration_days, price	1→N AuditRequest (chosen_by)
AUDIT_REQUEST	id (UUID), client_id (FK), pack_id (FK), status (pending/in_progress/done/archived), scope_notes, submitted_at, updated_at, completed_at	N→1 User+Pack ; 1→1 Report ; 1→N Notification
AUDIT_REPORT	id (UUID), audit_request_id (FK UK), pdf_path, summary, findings (JSON), generated_at	1→1 AuditRequest
TRAINING_MODULE	id, title, content_md, duration_min, level, published_at	1→N Progress
TRAINING_PROGRESS	id, user_id, module_id, completed, started_at, completed_at	N→1 User + Module
NOTIFICATION	id, user_id, request_id, type, subject, status, sent_at	N→1 User + Request

Recommended indexes:

- USER.email UNIQUE
- AUDIT_REQUEST.status BTREE (frequent admin filtering)
- (AUDIT_REQUEST.client_id, AUDIT_REQUEST.created_at) composite (client dashboard)
- AUDIT_REPORT.audit_request_id UNIQUE (1 report per request)

4.2 Back-end class diagram (Django models + services)

Class User

Represents a platform user, whether they are an SME client or an administrator. Manages JWT authentication and role separation.

Attribute	Description
id : UUID	Unique identifier (PK)
email : str	Email address (unique, indexed)
_password_hash : str	Hashed password (PBKDF2)
first_name, last_name	First and last name
role : enum	Client or admin
company_name : str	Company name (nullable)
is_active : bool	Account activated or deactivated
created_at : datetime	Account creation date

AuditPack Class

Represents one of the 4 packs (Audit, Security, Protection, Premium). Reference data loaded via luminaire.

Attribute	Description
id: int	Unique identifier
code: enum	audit / security / protection / premium
name: str	Commercial name
description: str	List of included services
duration_days: int	Incremental processing time
price: Decimal	price of the package

AuditRequest Class

Core of the business: the audit request submitted by an SME client.

Cycle of 4 statuses (pending → in_progress → completed → archived).

Attribute	Description
id: UUID	Unique identifier
reference: str	Case number (e.g., CYB-2026-0042)
client_id: UUID FK → User	SME Client who submitted
pack_id: UUID FK → AuditPack	Selected package
status: str(enum)	pending / in_progress / completed / archived
scope_notes: str	Form details
submitted_at : datetime	Cycle dates

Methods: generate_reference(), update_status(new_status), estimated_completion()

Class AuditReport

Vulnerability report linked to a single AuditRequest, includes a visual score and a downloadable PDF.

Attribute	Description
id: UUID	Unique identifier
audit_request_id: int FK	1-1 relationship
summary: str	Simplified executive summary
security_score: int	Score 0-100 (gauge)
findings: dict (JSON)	List of vulnerabilities
pdf_path: str	Path to generated PDF
generated_at : Datetime	Generation date

Methods: build_pdf(), get_download_url()

Class Notification

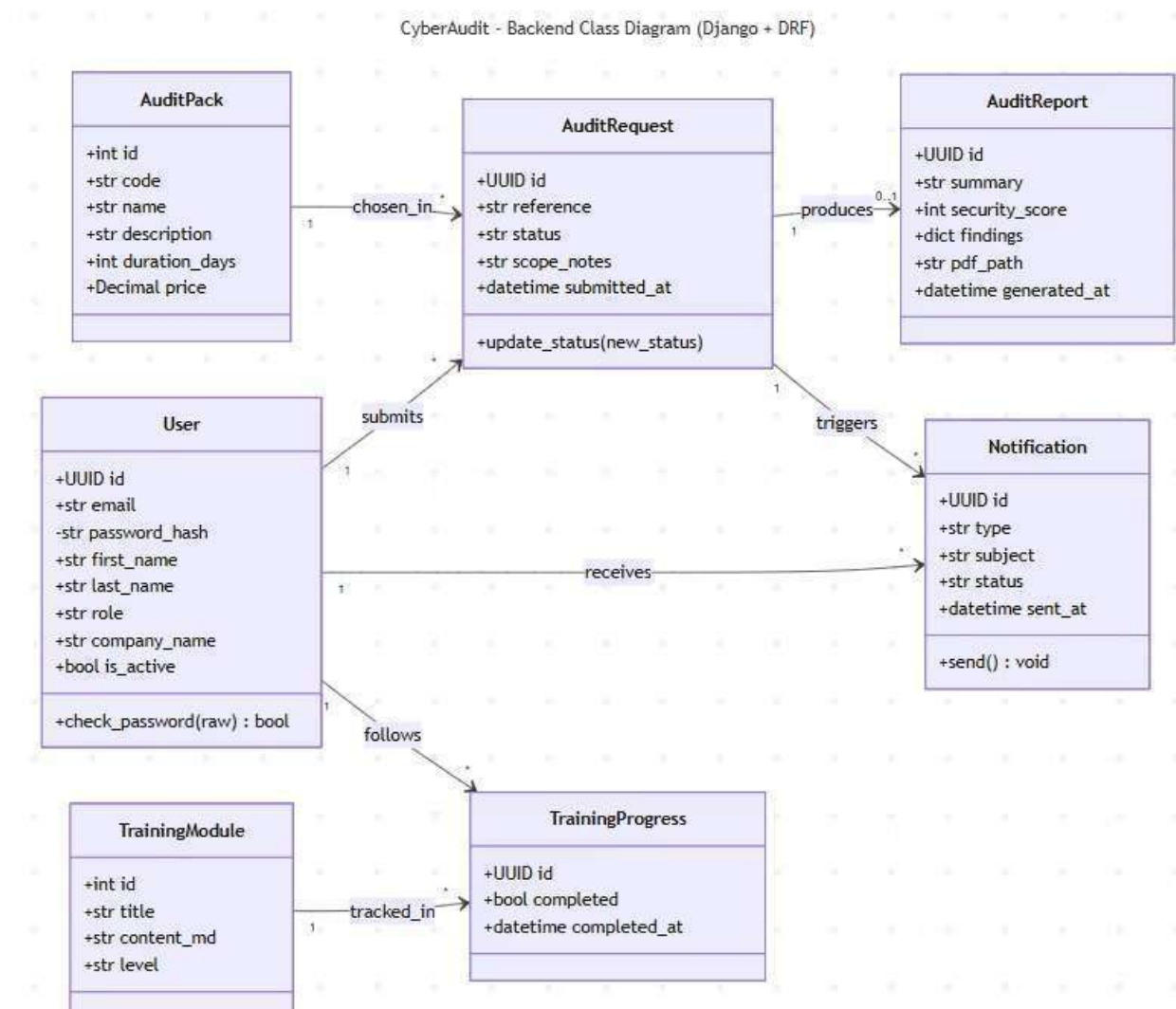
Trace toutes les communications e-mail envoyées via Celery.

Attribute	Description
id: UUID	Unique identifier
user_id: UUID FK → User	Recipient
request_id: UUID FK	Request concerned (nullable)
type: str(enum)	request_received / status_changed / report_ready
status: str(enum)	queued / sent / failed

Methods: send(), retry()

Classes TrainingModule & TrainingProgress

Cybersecurity awareness modules (phishing, MFA, GDPR...) with user progress tracking. Modules de sensibilisation cybersécurité (phishing, MFA, RGPD...) avec suivi de progression utilisateur.



4.3 Front-end components (React + Tailwind CSS)

The front-end is a React SPA communicating with the REST API via JWT. The site structure is organized into pages (full views), reusable UI components, a global store, and API calling services.

Authentication & routage

AuthStore : Global state (user, accessToken, refreshToken. Methods: login, logout, refresh, hasRole).

ProtectedRoute : Guard component, props (allowedRoles. Redirects to /login if not authenticated).

ApiClient : Axios wrapper that automatically adds the JWT to each request and handles the refresh token in case of a 401 error.

Public Pages

LoginPage : Login form. Submits to AuthStore.login() and then redirects according to the user's role.

RegisterPage : SME customer registration form. Calls AuthStore.register().

SME Customer Area

Client Dashboard: Dashboard listing the requests of the connected client.

AuditRequestForm: 3-step form (pack, scope, confirmation). Displays 4 Packcards.

RequestDetail: Detailed view with timeline, PDF report, and score.
Components: StatusBadge, SecurityScoreGauge, FindingsList.

CyberAudit Administrator Area

AdminDashboard: Paginated view of all requests with filters (status, package, date, client) and sorting.

Training module

TrainingPage: List of modules with progress indicator.

TrainingModuleViewer: Displays Markdown content. Security: cleanup via DOMPurify (antiXSS, SMART #3).

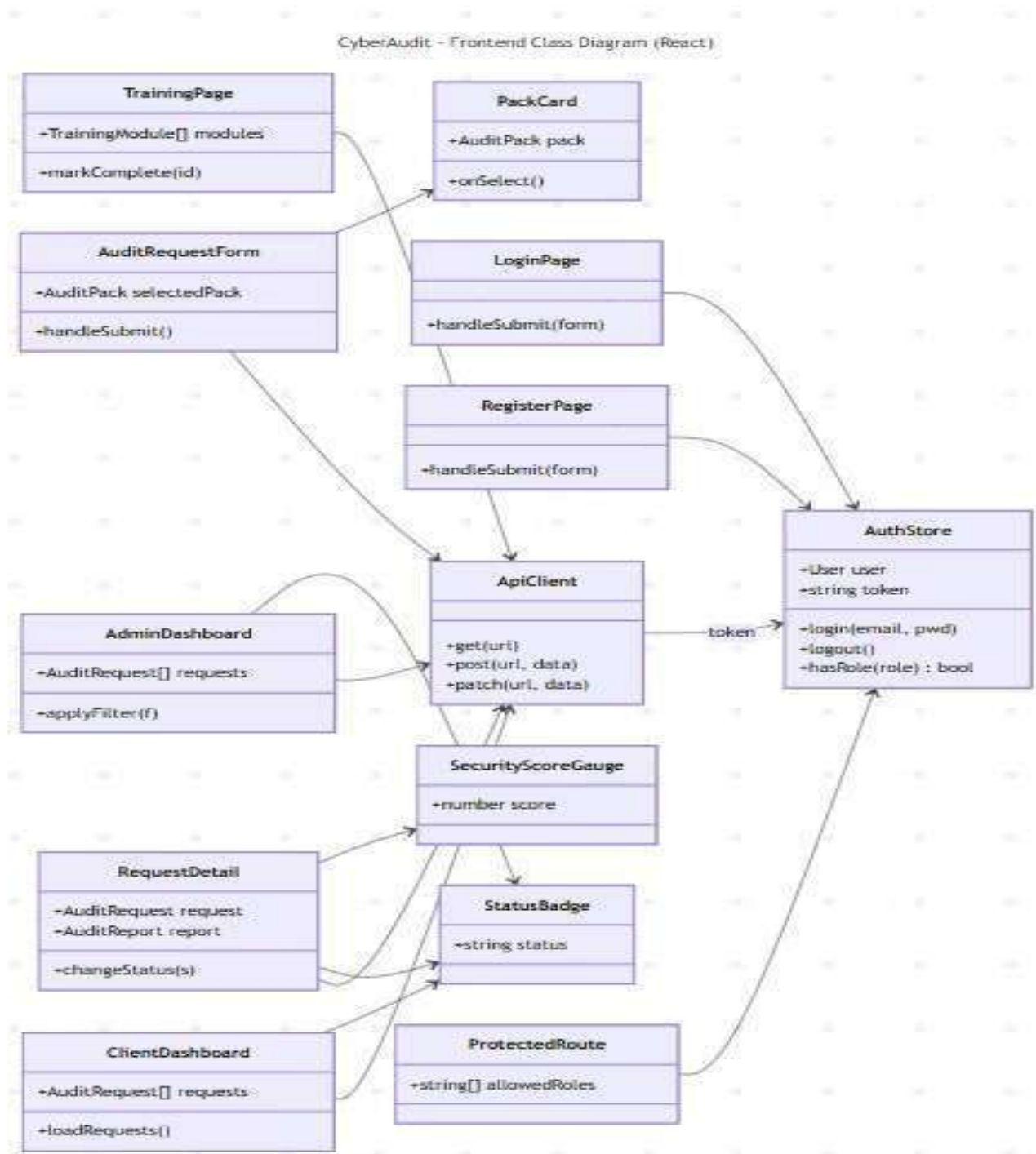
Reusable UI components

Navbar, **PackCard**, **StatusBadge** (grey/blue/green/purple), **SecurityScoreGauge** (circular gauge, red < 40, orange 40-70, green > 70), **RequestTable**, **Pagination**, **ToastContainer**.

Main Interaction Flow: Request Submission

1. The user logs in via LoginPage → AuthStore stores the JWT.
2. ProtectedRoute authorizes access to the ClientDashboard.
3. Clicking "New Request" → AuditRequestForm loads the packs.
4. Selecting a PackCard → AuditService.create() calls the Django API.
5. The backend creates the AuditRequest and triggers the Celery email task.
6. The client is redirected to RequestDetail (StatusBadge "pending").
7. A notification is sent for each status change on the admin side.

CyberAudit – Frontend Class Diagram (React) Explained



This diagram shows the React frontend architecture for a cybersecurity audit application called **CyberAudit**. It illustrates the main UI components (classes) and how they depend on shared services. Here's a walkthrough.

Central services

Two classes sit at the core of the application and are referenced by almost everything else: **ApiClient** is the HTTP layer. It exposes `get(url)`, `post(url, data)`, and `patch(url, data)` methods that the rest of the app uses to talk to the backend. It reads the auth token from **AuthStore** (shown by the labeled "token" arrow) so requests are authenticated automatically.

AuthStore holds session state: the current **User**, the JWT-style token, and methods `login(email, pwd)`, `logout()`, and `hasRole(role): bool`. Components consume it to know who's logged in and what they're allowed to do.

Authentication and routing

LoginPage and **RegisterPage** each expose a `handleSubmit(form)` method. They both depend on **ApiClient** (to send credentials to the backend) and on **AuthStore** (to persist the user/token after a successful response).

ProtectedRoute is a route guard with an `allowedRoles: string[]` property. It queries `AuthStore.hasRole(...)` to decide whether to render the child route or redirect — this is how the app enforces role-based access.

Client-facing screens

ClientDashboard holds `AuditRequest[]` requests and a `loadRequests()` method. It calls **ApiClient** to fetch the logged-in user's audit requests and uses **StatusBadge** to render their state.

AuditRequestForm lets a client submit a new audit. It carries a `selectedPack: AuditPack` and a `handleSubmit()` method and posts the form to the backend through **ApiClient**. It also renders **PackCard** components — each **PackCard** displays an **AuditPack** and exposes an `onSelect()` callback so the user can pick a pack.

RequestDetail is the per-request view. It holds an **AuditRequest** and an **AuditReport**, exposes `changeStatus(s)`, and composes two reusable widgets: **SecurityScoreGauge** (which displays a numeric score) and **StatusBadge** (which renders a status string). It talks to the backend via **ApiClient**.

TrainingPage shows `TrainingModule[]` with a `markComplete(id)` method — the training/awareness section of the app, also backed by **ApiClient**.

Admin side

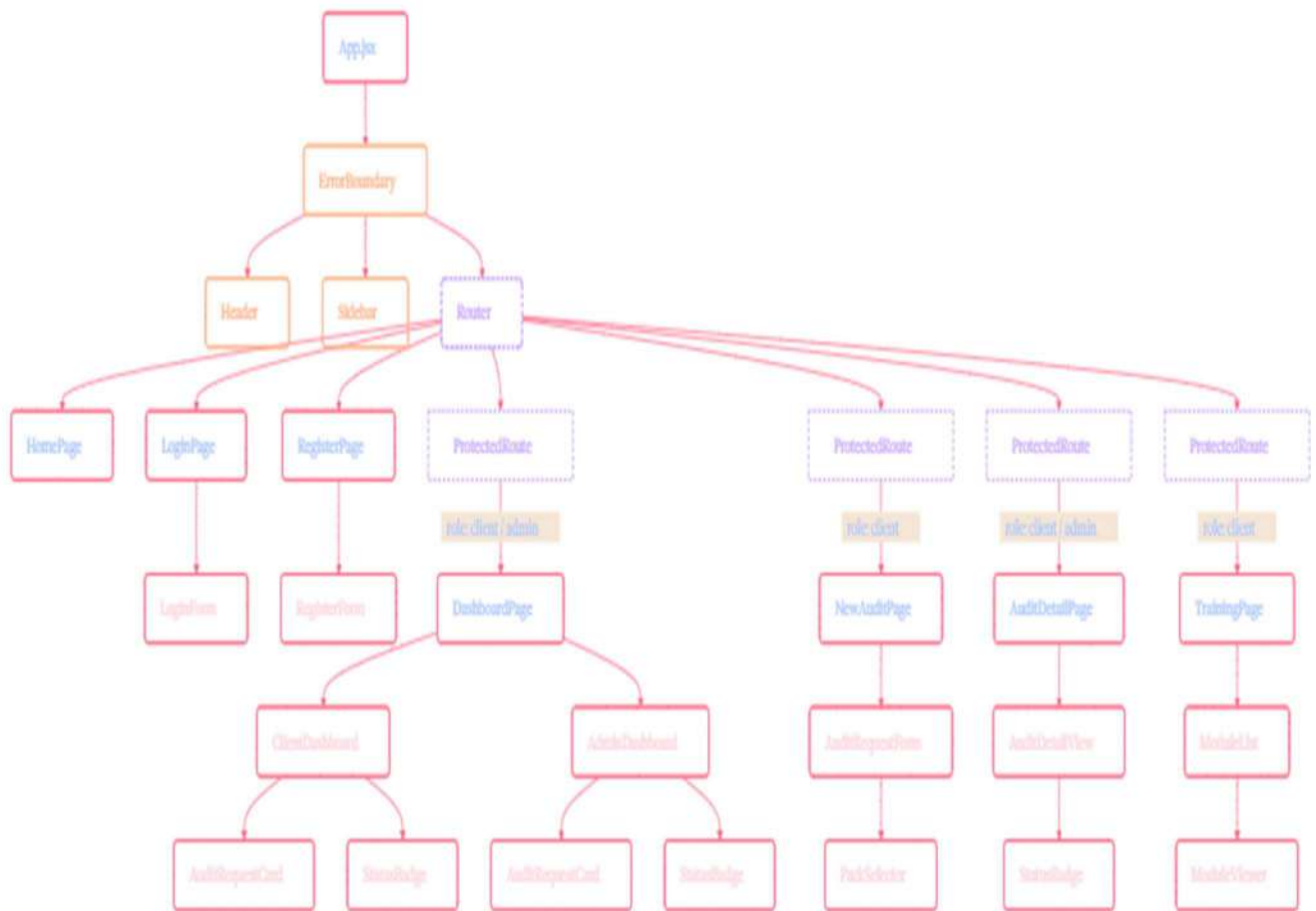
AdminDashboard mirrors the client dashboard but for administrators. It holds `AuditRequest[]` requests and an `applyFilter(f)` method, pulls data through **ApiClient**, and reuses **StatusBadge** for display.

How the pieces fit together

The architecture follows a clean layered pattern. Pages and dashboards (top layer) compose small reusable presentational components like **PackCard**, **StatusBadge**, and **SecurityScoreGauge**. They all go through a single **ApiClient** for backend communication, and a single **AuthStore** for identity and authorization. **ProtectedRoute** ties authorization into routing so unauthorized users never reach the protected pages in the first place.

In short: **AuthStore** = who you are, **ApiClient** = how you talk to the server, **ProtectedRoute** = what you're allowed to see, and everything else is feature-specific UI built on top of those three foundations.

Component tree



This diagram shows the component hierarchy of a web application, likely an audit platform with a training module. Read it top to bottom: each arrow means a component contains (or renders) another one.

At the top, **App.jsx** is the application's entry point. It wraps everything inside an **ErrorBoundary**, a component that catches JavaScript errors from its children so the whole app doesn't crash, a standard React best practice.

Below the **ErrorBoundary**, you have three structural components: the **Header** (top bar), the **Sidebar** (side menu), and the **Router**, which is the heart of navigation. **Header** and **Sidebar** are always visible, while the **Router** decides what to show in the main content area based on the URL.

The **Router** then dispatches to different pages:

First, the public pages, accessible to everyone: **HomePage** (landing page), **LoginPage** which contains a **LoginForm**, and **RegisterPage** which contains a **RegisterForm**.

Then, the protected pages, wrapped in **ProtectedRoute** components. This is a classic pattern:

ProtectedRoute checks that the user is authenticated and has the right role before rendering the page. There are four of them, each with different role restrictions:

- **DashboardPage** (client / admin roles) — renders either **ClientDashboard** or **AdminDashboard** depending on the role. Both reuse the same building blocks: **AuditRequestCard** and **StatusBadge**. A good example of reusable components.
- **NewAuditPage** (client role only) — contains an **AuditRequestForm** with a **FieldSelector** to choose the audit fields.
- **AuditDetailPage** (client / admin roles) — displays an **AuditDetailView** which also reuses the **StatusBadge**.
- **TrainingPage** (client role) — displays a **ModuleList** that renders **ModuleViewer** instances for each training module.

A few interesting things to note about the architecture:

Reusability is handled well: StatusBadge is used in three different places (ClientDashboard, AdminDashboard, AuditDetailView), and AuditRequestCard is shared between both dashboards. That's a sign of good decomposition.

Separation of concerns is clear: the pages (*Page) handle routing and permissions, while the views (*View, *Dashboard, *Form) handle rendering and business logic.

Access control is centralized in ProtectedRoute rather than scattered across each page, which makes security much easier to audit.

src/	
├ components/	
│ └ auth/	LoginForm, RegisterForm, ProtectedRoute
│ └ dashboard/	ClientDashboard, AdminDashboard, AuditRequestCard, StatusBadge
│ └ audit/	AuditRequestForm, PackSelector, AuditDetailView
│ └ training/	ModuleList, ModuleViewer
│ └ shared/	Header, Sidebar, Button, ErrorBoundary
├ pages/	HomePage, LoginPage, RegisterPage, DashboardPage, NewAuditPage, AuditDetailPage, TrainingPage
├ hooks/	useAuth, useAudits, useApi
├ services/	api.js (axios + JWT), auth/audits/reports services
├ utils/	sanitize.js (DOMPurify), validators.js
└ App.jsx	routing global

Conventions: Functional components + hooks (no classes), global state via Context API, sanitize() on all inputs.

4.4 Generation of the PDF vulnerability report

The PDF report is the SME client's main deliverable: it consolidates the vulnerability analysis, the security score, and the recommended action plan. Its generation is entirely asynchronous to avoid blocking the admin interface.

Stack technique

- **WeasyPrint:** Converts an HTML+CSS template into a print-quality PDF. Reuses the web's style guide.
- **Celery:** Runs the `generate_pdf_task` in the background.
- **Redis:** Celery queue (parallel scaling).
- **Stockage:** `/media/reports/` in MVP, S3 in V2; access via signed URL (15 min).

Structure of the generated PDF

Section	Contents
Page 1: Cover	CyberAudit logo, client name, case reference (CYB-2026-0042), date, confidentiality
Page 2: Executive Summary	Overall score A to F (scale 0-100) + summary in 3 simplified lines
Page 3: Audited Scope	List of assets analyzed (mail server, Wi-Fi router, workstations, etc.) + EBIOS RM methodology
Pages 4-N: Vulnerabilities	Table findings: severity (Critical/High/Medium/Low), active, description, recommendation
Final page : 30-day action plan	Weekly steps (S1, S2, S3, S4) with concrete actions and priorities
Appendix	Glossary, references (ISO 27005, ANSSI), contact CyberAudit

Full Flow — From Request to Download

1. The admin clicks **Generate PDF report** in AdminRequestDetail.
2. The front calls POST `/api/requests/{id}/generate-report` with findings.
3. The PdfReportService (Django) creates the AuditReport (summary, score, findings JSON).
4. It dispatches `generate_pdf_task.delay(report_id)` and returns 202 Accepted.
5. The Celery worker renders `report.html` with the data; WeasyPrint produces the PDF.
6. File saved to `/media/reports/CYB-2026-0042.pdf`, `pdf_path` updated.
7. A `report_ready` Notification is created, email sent via `send_email_task`.
8. Client-side, ClientDashboard shows **Download report** → GET `/api/reports/{id}/download` streaming.

PDF Access Security

- **JWT required** on `/api/reports/{id}/download`. No anonymous access.
- **IsAdminOrOwner permission** : 403 Forbidden otherwise.
- **Signed URL** with timestamp, expires after 15 minutes.
- **No browser cache** : Cache-Control, no-store, private.

- **Audit log** (V2) : who, when, from which IP, GDPR-compliant traceability.

Generation Failure Handling

- **Auto-retry** : autoretry_for=(Exception,), max_retries=3, exponential backoff (10s, 30s, 90s).
- **Intermediate status** : pdf_path empty until generated → front shows "Generating...".
- **Failure notification** : after 3 failures, generation_failed notification sent to admin.
- **Manual regeneration** via the **Generate PDF report** button.

4.5 Monitoring & Observability

The application must remain traceable in production: we must know *who did what, when, and why it crashed*. The strategy rests on 3 pillars: structured logs, error monitoring, alerting.

4.5.1 Structured Logs (JSON)

Django configures structlog to produce JSON logs consumable by Railway and a future ELK/Loki stack. Each log carries at minimum: timestamp, level, request_id (correlation), user_id, endpoint, latency_ms.

```
{
  "timestamp": "2026-05-18T09:24:11Z",
  "level": "INFO",
  "event": "audit_request.created",
  "request_id": "req-7f2a...",
  "user_id": "8f2a...c9",
  "audit_id": "CYB-2026-0042",
  "pack": "security",
  "latency_ms": 142
}
```

4.5.2 Error Monitoring : Sentry

- **Back-end**: sentry-sdk[30django,celery] captures all Django and Celery exceptions with stack trace, user context, request payload (PII anonymized).
- **Front-end**: @sentry/react captures JS errors and unhandled promise rejections, plus session replay on error.
- **Release tracking**: every git tag pushes a Sentry release → per-version regression tracking.
- **Sample rate**: 100% of errors on MVP; 10% of performance transactions to stay within the free quota.

4.5.3 Business & Technical Metrics

Metric	Source	Alert Threshold
API response time (p95)	Sentry Performance / Railway metrics	> 800 ms for 5 min
5xx error rate	Sentry	> 1% over 10 min
Consecutive Celery failures	Structured logs	≥ 3 on the same task
Active DB connections	Railway Postgres metrics	> 80% of pool
/media disk space	Railway	> 85%
Audit requests / day	Django + admin dashboard	Growth metric, no alert

4.5.4 Alerting

- **Sentry** emails the tech lead on every new unseen error, and a Slack DM (V2) if error rate > threshold.
- **Healthcheck**: GET /api/health/ endpoint checks DB, Redis, SMTP. Monitored by UptimeRobot (free) every 5 min.
- **Celery failure notification**: after 3 retries, an admin email is sent (already documented in §4.4).

4.5.5 Application Audit Log (V2)

An AUDIT_LOG table tracks sensitive actions: logins, report downloads, status changes, account deletions. Fields: id, user_id, action, resource, ip, user_agent, created_at. Retention 1 year, GDPR-compliant (see §7.5).

4.6 Database Migrations & Seeding

Schema version transitions and pre-population of reference data (4 audit packs, initial training modules, admin account) are fully automated. No manual intervention in production.

4.6.1 Django Migrations Strategy

- **Convention**: 1 migration = 1 atomic change. Explicit name: 0007_add_security_score_to_auditreport.py.
- **Schema migrations**: generated via python manage.py makemigrations, never hand-edited.
- **Data migrations**: use migrations.RunPython(forward, reverse) with a systematic reverse function to allow rollback.
- **CI enforcement**: python manage.py migrate --check in the pipeline to detect non-generated migrations.
- **Production**: Railway runs migrate automatically on deploy via the Procfile release command.

4.6.2 Example Data Migration

```
# audits/migrations/0002_seed_packs.py
from django.db import migrations

def seed_packs(apps, schema_editor):
    AuditPack = apps.get_model('audits', 'AuditPack')
    AuditPack.objects.bulk_create([
        AuditPack(code='audit', name='Audit', duration_days=5, price=490),
        AuditPack(code='security', name='Security', duration_days=10, price=990),
        AuditPack(code='protection', name='Protection', duration_days=15, price=1490),
        AuditPack(code='premium', name='Premium', duration_days=20, price=2490),
    ])

def remove_packs(apps, schema_editor):
    apps.get_model('audits', 'AuditPack').objects.all().delete()

class Migration(migrations.Migration):
    dependencies = [('audits', '0001_initial')]
    operations = [migrations.RunPython(seed_packs, remove_packs)]
```

4.6.3 Fixtures (Demo Data)

JSON fixtures in backend/fixtures/ load demo data (5 fake clients, 10 audit requests, 3 reports) for jury demos and E2E tests:

```
python manage.py loaddata fixtures/demo_clients.json
python manage.py loaddata fixtures/demo_audits.json
python manage.py loaddata fixtures/demo_reports.json
```

A wrapper script scripts/seed_demo.sh chains migrate + loaddata for a 1-command demo environment.

4.6.4 Initial Admin Creation

No plaintext passwords in code. A custom Django command create_initial_admin reads ADMIN_EMAIL and ADMIN_PASSWORD from environment variables and creates the user if the table is empty:

```
python manage.py create_initial_admin
# Idempotent: does nothing if an admin already exists
```

4.6.5 Rollback Strategy

Scenario	Procedure
Broken migration just after deploying	python manage.py migrate audits 0006 reverts to the previous migration. The RunPython reverse function executes.
Corrupted data detected	Restore the latest Railway backup (1-click), then investigate off-prod on a snapshot.
Migration requiring downtime	Planned maintenance window, Vercel maintenance page enabled, migration executed, verification, reopening.
Non-backward-compatible schema change	"Expand & contract" strategy: 1) add new column, 2) dual-write in app, 3) backfill, 4) switch reads, 5) drop old column

4.6.6 Snapshot Before Each Release

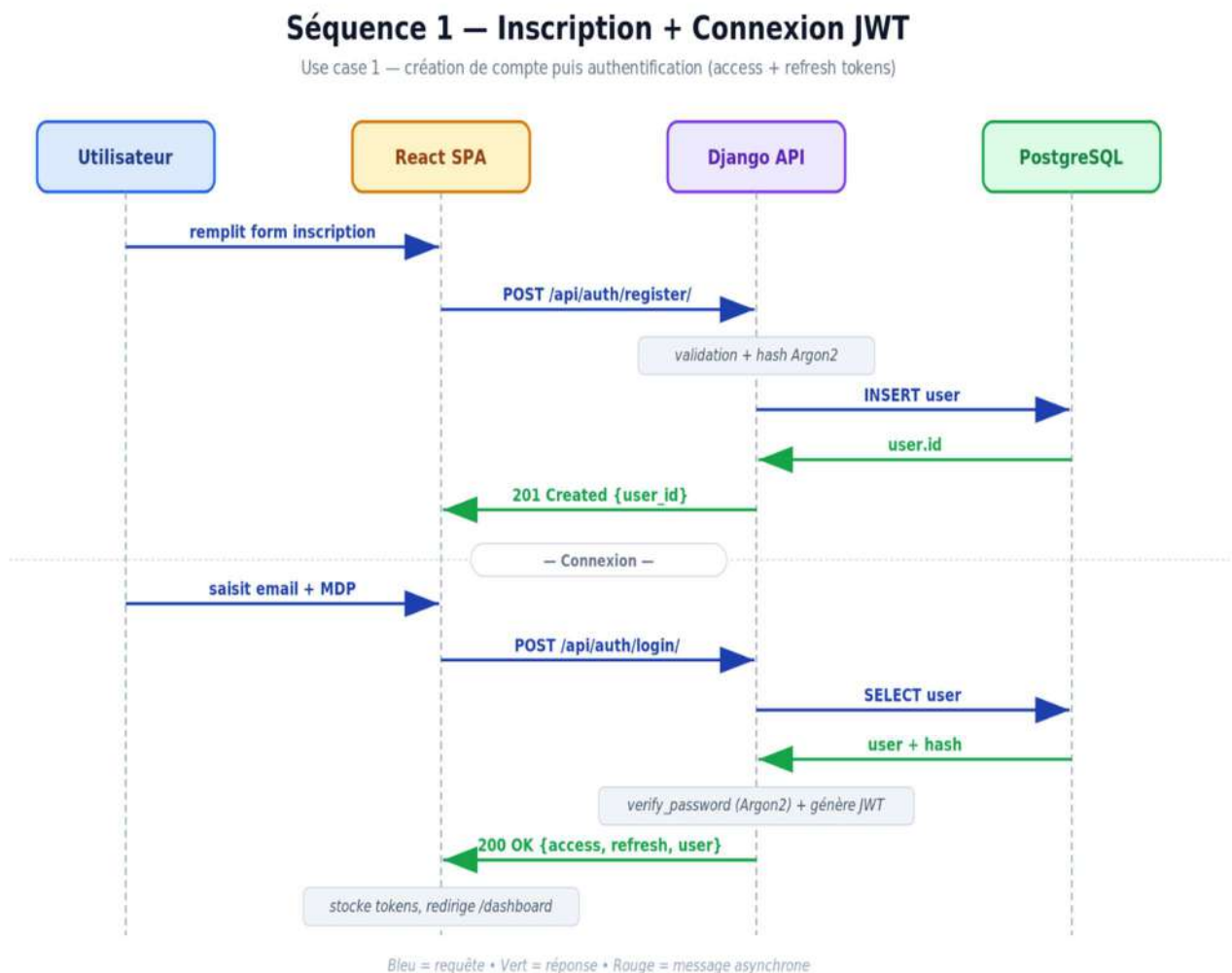
Before each merge to main (prod release), a manual Railway DB snapshot is triggered via railway db snapshot create. 14-day retention. Allows rollback in < 5 min in case of major issue.

Sequence Diagrams

5.1 Sign-up + JWT Login

The user enters email and password. The front calls POST /api/auth/register/, then POST /api/auth/login/. Django validates credentials, issues access + refresh tokens and returns them. The front stores them in AuthStore and redirects to the role-appropriate dashboard.

Figure 5 : Sign-up + JWT Login Sequence



This diagram illustrates **Use Case 1: Registration then Login** for a web application, using JWT (JSON Web Tokens) for authentication. It shows how four actors interact over time: the **User**, the **React SPA** (Single Page Application, the frontend), the **Django API** (the backend), and the **PostgreSQL** database.

Messages Exchanged (Detailed)

Marie → **RegisterPage**: fills out the form (email, password, company_name).
RegisterPage → **ApiClient**: POST /api/auth/register/ with JSON body {email, password, company_name, company_size, sector}.
ApiClient → **Django (RegisterView)**: DRF serializer validates fields (unique email, password strength).
Django → **PostgreSQL**: INSERT INTO accounts_user with bcrypt-hashed password.
Django → **Celery**: send_welcome_email.delay(user_id).
Django → **RegisterPage**: 201 Created with {user, access, refresh}.
RegisterPage → **AuthStore**: stores accessToken and refreshToken.
RegisterPage → **React Router**: redirect to /dashboard based on user.role.
Celery worker → **SMTP Gmail**: sends welcome email (asynchronous, off the HTTP thread).

Mermaid (renders on GitHub)

```
sequenceDiagram
```

```
actor Marie
participant Front as RegisterPage
participant API as Django /auth
participant DB as PostgreSQL
participant Celery
participant SMTP as Gmail SMTP
```

```
Marie->>Front: Fill form
Front->>API: POST /auth/register/ {email, password, ...}
API->>API: Validate (DRF serializer)
API->>DB: INSERT user (bcrypt password)
API->>Celery: send_welcome_email.delay(user_id)
API-->>Front: 201 {user, access, refresh}
Front->>Front: AuthStore.set(tokens)
Front-->>Marie: Redirect /dashboard
Celery->>SMTP: Welcome email
SMTP-->>Marie: Email received
```

5.2 Audit Request Submission (async)

The client submits the audit form. The front calls POST /api/audits/ with the JWT in headers. Django validates via the DRF serializer, persists to the database, enqueues a Celery acknowledgment-email task, and responds 201 Created. The Celery worker pops the task from Redis and sends the email via SMTP.

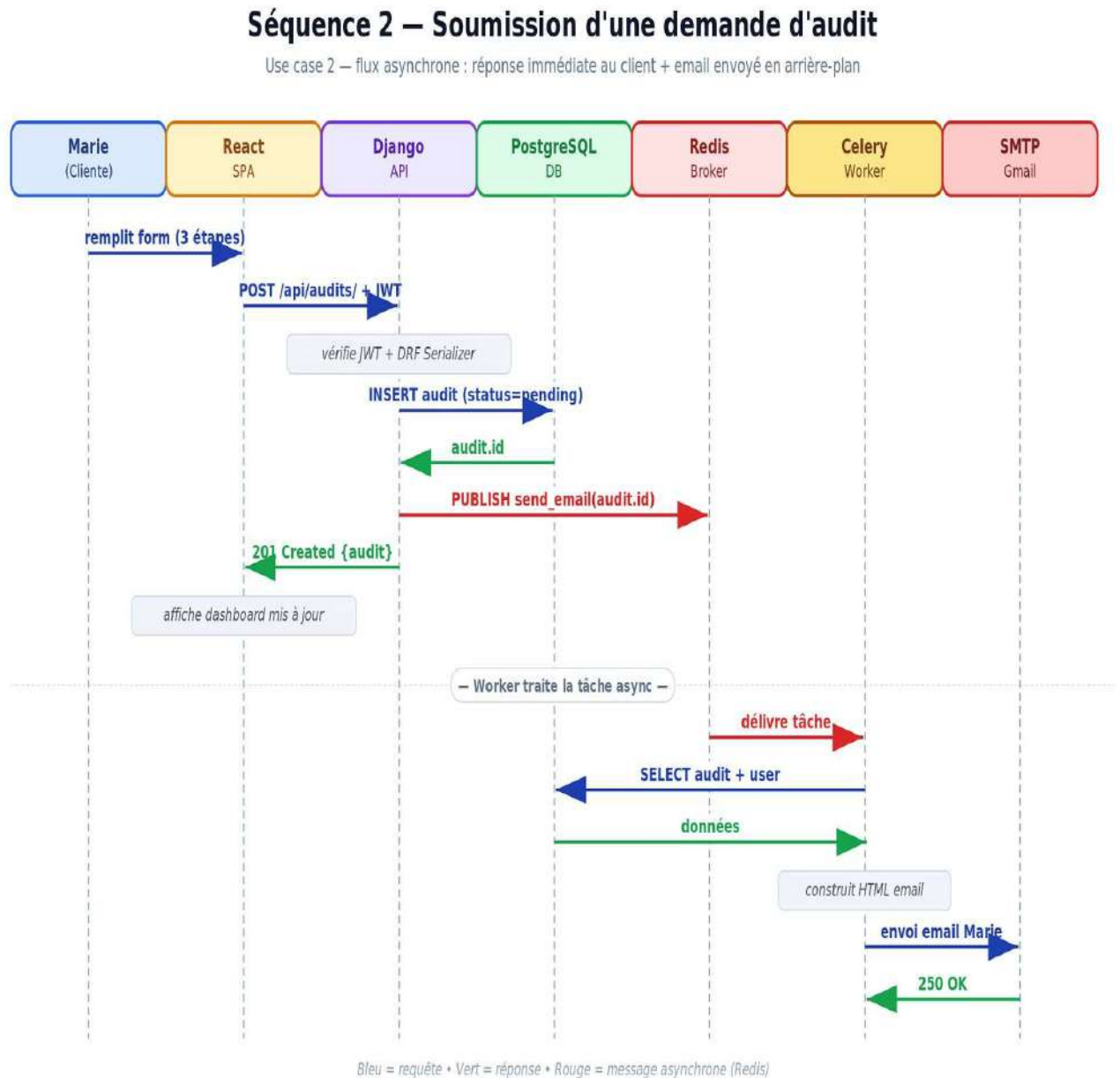


Figure 6: Async Audit Submission Sequence

Messages Exchanged (Detailed)

Marie → AuditRequestForm: selects a pack and enters scope_notes.

AuditRequestForm → ApiClient: POST /api/audits/, header Authorization: Bearer <JWT>, body {pack_id, scope_notes}.

ApiClient → Django (AuditCreateView): middleware checks JWT, DRF validates fields.

Django → PostgreSQL: INSERT INTO audits_auditrequest with status='pending', reference='CYB-2026-XXXX'.

Django → Redis (Celery broker): enqueues send_acknowledgment_email.delay(audit_id).

Django → AuditRequestForm: 201 Created with request details.

AuditRequestForm → ClientDashboard: redirect with "Request sent" toast.

Celery worker ← Redis: dequeues the task.

Celery worker → SMTP Gmail: sends acknowledgment to Marie + notification to Karim.

Celery worker → PostgreSQL: UPDATE notifications SET status='sent'.

Mermaid (renders on GitHub)

```
sequenceDiagram
```

```

actor Marie
participant Front as AuditRequestForm
participant API as Django /audits
participant DB as PostgreSQL
participant Redis
participant Worker as Celery Worker
participant SMTP

Marie->>Front: Choose pack + scope_notes
Front->>API: POST /audits/ + JWT
API->>API: Verify JWT + validate
API->>DB: INSERT audit_request (status=pending)
API->>Redis: enqueue send_ack_email(audit_id)
API-->>Front: 201 Created {audit}
Front-->>Marie: Redirect dashboard + toast
Worker->>Redis: dequeue task
Worker->>SMTP: Acknowledgment email
Worker->>DB: UPDATE notification status=sent
SMTP-->>Marie: Email received

```

5.3 PDF Report Generation (admin)

The admin clicks **Generate PDF report**. The front calls POST /api/audits/{id}/generate-report/. Django creates the AuditReport and enqueues the Celery task, returning 202 Accepted. The Celery worker renders the HTML template, WeasyPrint produces the PDF, the file is stored, and an email notification is sent to the client.

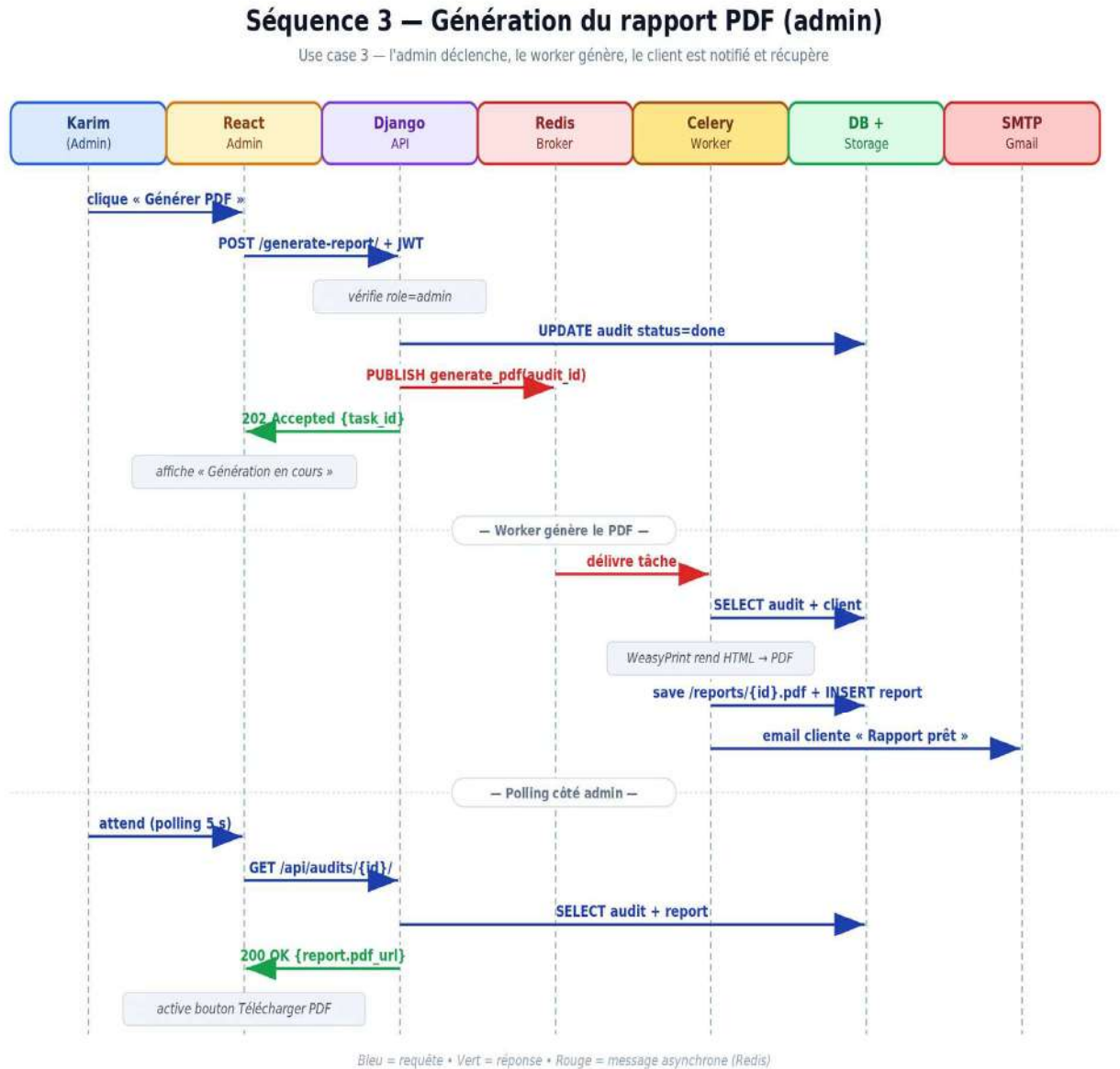


Figure 7: PDF Report Generation Sequence

Messages Exchanged (Detailed)

Karim → AdminRequestDetail: clicks "Generate PDF report" with entered findings.

AdminRequestDetail → ApiClient: POST /api/audits/{id}/generate-report/, body {findings, summary, security_score}.

ApiClient → Django (PdfReportService): checks IsAdmin permission, verifies status='done'.

Django → PostgreSQL: INSERT INTO reports_auditreport with pdf_path=NULL (in progress).

Django → Redis: enqueues generate_pdf_task.delay(report_id).

Django → AdminRequestDetail: 202 Accepted with {report_id, status: "generating"}.

Celery worker ← Redis: dequeues the task.

Celery worker → Django templates: renders report.html with context (findings, score, recommendations).

Celery worker → WeasyPrint: converts HTML+CSS to binary PDF.

Celery worker → Storage: saves to /media/reports/CYB-2026-0042.pdf.

Celery worker → PostgreSQL: UPDATE auditreport SET pdf_path=....

Celery worker → SMTP Gmail: sends "Report available" email to Marie.

Marie → ClientDashboard: sees "Download report" link appear (30 s polling).

Marie → Django: GET /api/audits/{id}/report/ with JWT.

Django → Marie: 15-min signed URL, PDF streaming.

Mermaid (renders on GitHub)

```
sequenceDiagram
    actor Karim
    participant Front as AdminRequestDetail
    participant API as Django /audits
    participant DB as PostgreSQL
    participant Redis
    participant Worker as Celery Worker
    participant WP as WeasyPrint
    participant Storage
    participant SMTP
    actor Marie

    Karim->>Front: Click Generate PDF report
    Front->>API: POST /audits/{id}/generate-report/
    API->>DB: INSERT auditreport (pdf_path=NULL)
    API->>Redis: enqueue generate_pdf_task
    API-->>Front: 202 Accepted
    Worker->>Redis: dequeue
    Worker->>Worker: render report.html
    Worker->>WP: HTML+CSS to PDF
    WP-->>Worker: PDF bytes
    Worker->>Storage: save /media/reports/CYB-...pdf
    Worker->>DB: UPDATE pdf_path
    Worker->>SMTP: Email Report available
    SMTP-->>Marie: Email received
    Marie->>API: GET /audits/{id}/report/ + JWT
    API-->>Marie: Stream PDF (signed URL 15min)
```

API Documentation

6.1 External APIs Consumed

External API	Usage	Reason
SMTP Gmail	Transactional emails (sign-up confirmation, status change, report delivery)	Free up to 500 emails/day, simple, reliable. Migration to SendGrid/Mailgun possible in V2.
Let's Encrypt	Automatic HTTPS certificates	Free, natively integrated in Vercel and Railway.

6.2 Internal API : REST Endpoints

Convention: prefix /api/, JSON for requests and responses, JWT in Authorization: Bearer <token>, standard HTTP codes.

6.2.1 Authentication : /api/auth/

Method	URL	Auth	Description	Body	2xx Response	Errors
POST	/api/auth/register/	—	Create client account	{email, password, company_name, company_size, sector}	201 {user, access, refresh}	400, 409 email exists
POST	/api/auth/login/	—	Login + JWT tokens	{email, password}	200 {access, refresh, user}	401, 429 rate limit
POST	/api/auth/refresh/	—	Refresh access token	{refresh}	200 {access}	401 refresh expired
POST	/api/auth/logout/	✓	Invalidate refresh	{refresh}	204	401
POST	/api/auth/password-reset/	—	Send reset email	{email}	200 (always, anti-leak)	—

6.2.2 Audit Requests : /api/audits/

Method	URL	Auth	Role	Describe
GET	/api/audits/	✓	client / admin	Paginated list (client → own / admin → all)
POST	/api/audits/	✓	client	Create a new request
GET	/api/audits/{id}/	✓	owner / admin	Request detail
PATCH	/api/audits/{id}/	✓	admin	Edit status or notes
DELETE	/api/audits/{id}/	✓	admin	Archive (soft delete)
POST	/api/audits/{id}/generate-report/	✓	admin	Trigger PDF generation (Celery)
GET	/api/audits/{id}/report/	✓	owner / admin	Download the PDF (binary)

Example ; Creating a Request

```
POST /api/audits/
Authorization: Bearer eyJhbGc...
Content-Type: application/json
```

```
{ "pack_id": 2, "scope_notes": "Audit of file server + 12 Windows workstations." }
```

```
HTTP/1.1 201 Created
Content-Type: application/json
```

```
{
  "id": "8f2a...c9",
  "client": "user-uuid",
  "pack": {"id": 2, "name": "standard"},
  "status": "pending",
  "scope_notes": "Audit of file server ...",
  "created_at": "2026-05-18T09:24:11Z"
}
```

6.2.3 Audit Packs : /api/packs/

Method	URL	Auth	Description
GET	/api/packs/	—	Public list of the 4 packs
GET	/api/packs/{id}/	—	Pack detail

6.2.4 Training : /api/training/

Method	URL	Auth	Description
GET	/api/training/modules/	✓	List available modules
GET	/api/training/modules/{id}/	✓	Module content
POST	/api/training/modules/{id}/start/	✓	Mark as started
POST	/api/training/modules/{id}/complete/	✓	Mark as completed

6.2.5 User Profile : /api/users/

Method	URL	Auth	Description
GET	/api/users/me/	✓	Current user's profile
PATCH	/api/users/me/	✓	Update own profile
POST	/api/users/me/change-password/	✓	Change own password
DELETE	/api/users/me/	✓	Delete own account (GDPR)

6.3 Cross-Cutting Conventions

- **Pagination** : ?page=1&page_size=20, response {count, next, previous, results}
- **Filtering** : ?status=pending&ordering=-created_at
- **Errors** : unified format {error: "code", message: "Message", details: {...}}
- **Rate limiting** : 5 req/min on /auth/login/ and /auth/password-reset/ (IP-based)
- **Versioning** : no v1 in URL on the MVP; in V2 if breaking changes → /api/v2/

SCM & QA Strategy

7.1 Source Code Management (Git)

Host: GitHub repo “Tommy-JOUHANS/portfolio” (private until defense, then public).

Branching Strategy (lightweight Git Flow)

main ← production only (protected, Vercel/Railway continuous deploy)
 develop ← integration branch (CI required)
 feature/<name> ← new features (1 PR = 1 feature)
 fix/<name> ← bug fixes
 hotfix/<name> ← urgent fixes directly on main
 release/<v> ← stabilization before release (from W11)

Commit Convention : Conventional Commits

feat(audits): add request form
 fix(auth): fix expired JWT refresh token
 docs(readme): update technical stack
 test(audits): add CRUD unit tests
 chore(deps): bump Django 5.0.2 → 5.0.3
 refactor(reports): extract PDFGenerator service

Pull Request Rules

Rule	Detail
Required review	Each PR validated by the other member before merge
Green CI	Tests + lint OK before merge
No direct push	main and develop protected by GitHub Branch Protection
Structured description	"What changes / Why / How to test"
Squash merge	1 PR = 1 commit in develop (clean history)

7.2 QA Strategy

Test Pyramid

```

      /\
     /E2E\          5 % - Cypress (critical journeys)
    /-----\
   /  Integ. \ 20 % - API + DB integration (Django + DRF)
  /-----\
 /  Unit tests \ 75 % - Jest (front) + pytest (back)
/-----\

```

Chosen Tools

Layer	Tool	Target
Back-end unit	pytest + pytest-django	Models, services, serializers
Back-end API	pytest + APIClient (DRF)	auth, audits, reports endpoints
Front-end unit	Jest + React Testing Library	Components, hooks, utils
E2E	Cypress	Sign-up → request → email (mock)
Security	Postman (collection) + bandit	XSS, SQLi, brute-force, JWT tampering
UX	3 user sessions (W7-W10)	Marie persona, < 3-min flow
Coverage	coverage.py + jest --coverage	Target: 80% min on critical apps
Lint	ruff + black ; eslint + prettier	Pre-commit hook
Type checking	mypy ; propTypes or TS (V2)	From W4

7.3 Manual Security Tests (W7-W10)

Test	Method	Expected Result
Persistent XSS	<script>alert(1)</script> in scope_notes	Escaped / sanitized
SQL injection	' OR 1=1 -- in login email	No leak, ORM bind
Brute-force	100 login attempts	Blocked from 6th (rate-limit)
JWT tampering	Modify role in payload	Invalid signature → 401
CSRF	POST without Authorization header	401
HSTS / CSP	HTTP headers in prod	Present and strict

7.4 CI/CD Pipeline

GitHub Actions

```
# .github/workflows/ci.yml
name: CI
on: [push, pull_request]

jobs:
  backend:
    runs-on: ubuntu-latest
    steps:
      - checkout
      - setup-python 3.11
      - install: pip install -r backend/requirements-dev.txt
      - lint: ruff check . && black --check .
      - test: pytest --cov=apps --cov-fail-under=80

  frontend:
    runs-on: ubuntu-latest
    steps:
      - checkout
      - setup-node 18
      - install: npm ci
      - lint: npm run lint
      - test: npm test -- --coverage
      - build: npm run build
```

Continuous deployment:

Environment	Trigger	Target	URL
Preview	All PR	Vercel preview + Railway branch deployment	pr-XX.preview.cyberaudit.app
Staging	Merge on develop	Vercel staging + Railway staging	staging.cyberaudit.app
Production	Merge on main (release tag)	Vercel + Railway prod	cyberaudit.app

Release strategy:

- SemVer versioning: 0.1.0 (W4 partial MVP) → 0.5.0 (W6 async) → 1.0.0 (W12 presentation)
- Annotated git tag for each release: `git tag -a v1.0.0 -m "MVP presentation Holberton"`
- Changelog kept up to date in CHANGELOG.md

Appendices

8.1 Glossaire

Term	Definition
MVP	Minimum Viable Product : minimal deliverable version of the product
JWT	Web Token : stateless authentication standard
MoSCoW	Prioritization method: Must / Should / Could / Won't
CRUD	CRUD Create / Read / Update / Delete
ORM	ORM Object Relational Mapping : database abstraction
CI/CD	CI/CD Continuous Integration / Continuous Deployment
GDPR	GDPR General Data Protection Regulation
HSTS HTTP	Strict Transport Security : anti-downgrade header
CSP	Content Security Policy : anti-XSS header
SemVer	Semantic Versioning : MAJOR.MINOR.PATCH
SPA	Single Page Application

8.2 Mapping Stage 3 ↔ RNCP 5 skills

RNCP Web and Mobile Web Developer (TP issued by the Ministry of Labor). This document covers the following blocks:

RNCP Block	Skill	Documentation section
BC01 : Develop the front-end part	Mock up an application	§2.3 (mockups)
BC01	Create static and dynamic interfaces	§4.3 (Component Responses)
BC01	Develop the front-end part	§3 (Architecture), §6 (API)
BC02 : Develop the back-end part	Create a database	§4.1 (Entity Relationship diagram)
BC02	Develop data access components	§4.2 (Django Template)
BC02	Develop the back-end part	§6 (REST API)
BC02	Design and implement components in a content management or e-commerce application	§4.2 (Services), §5 (Sequences)
Transversal	Work in an agile team / Versioning	§7 (SCM/QA)

8.3 Technical sources & references

Reference	Links
Django REST Framework : Documentation	https://www.django-rest-framework.org
SimpleJWT : JWT for Django	https://django-rest-framework-simplejwt.readthedocs.io
Celery : Distributed Task Queue	https://docs.celeryq.dev
WeasyPrint : HTML to PDF	https://weasyprint.org
OWASP Top 10 :security repository	https://owasp.org/www-project-top-ten
Conventional Commits	https://www.conventionalcommits.org
RNCP 38038 : Web and Mobile Web Developer	https://www.francecompetences.fr/recherche/rncp/38038
ANSSI : Overview of the cyber threat 2023	https://www.ssi.gouv.fr

8.4 Internal links within the project

- Stage 1: Project Charter (FR)
- Stage 2: Planning & Gantt (FR)
- Figma Wireframes
- MVP Code Repo

Document prepared as part of Stage 3 : Technical Documentation, Holberton School Dijon, class of 2026. Designed to be reused as-is in the RNCP 5 certification folder “Web and Mobile Web Developer”.